

Can Breast MRI Help Predict Treatment Response?

What vessel networks taught us—and what they didn't

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Before surgery, chemotherapy is a high-stakes test

THE TREATMENT

Before surgery

Chemotherapy can shrink a tumor and reveal how it responds.

THE OUTCOME

≈ 1 in 3

scans came from patients with no remaining invasive cancer at surgery.

THE QUESTION

Can MRI help?

Could an existing scan give us an earlier clue about treatment response?

The long-term goal is better evidence for tailoring treatment—not replacing clinical judgment.

An MRI “movie” shows more than anatomy

After contrast is injected, repeated MRI frames reveal **how it appears through the breast’s vessels over time.**



179

analyzed exams

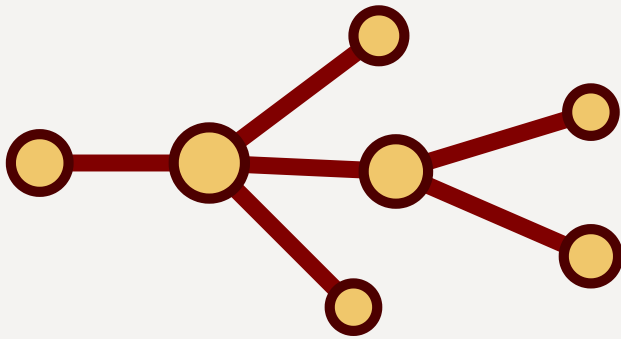
143

patients

13–24

frames per MRI movie

We asked whether the vessel map adds value



Graph neural network

Each vessel location shares information with its neighbors.

vs.

How bright?

How fast?

How long?

Simple summary

Average the same vessel measurements across the whole scan.

Does knowing **which vessel connects to which** improve the prediction?

Vessel behavior carried a modest signal

Ranking score: 50 = chance · 100 = perfect



Something is there.

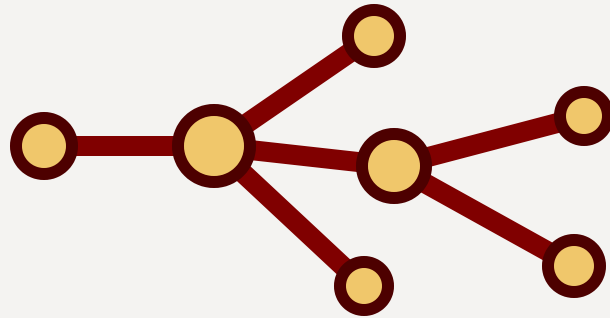
Vessel enhancement ranked responders above non-responders about 61 times out of 100.

But the estimate is uncertain.

Differences in scan timing may explain part of it; the defensible range is roughly **54–61**.

The clearest result came from deleting the graph

Real vessel connections

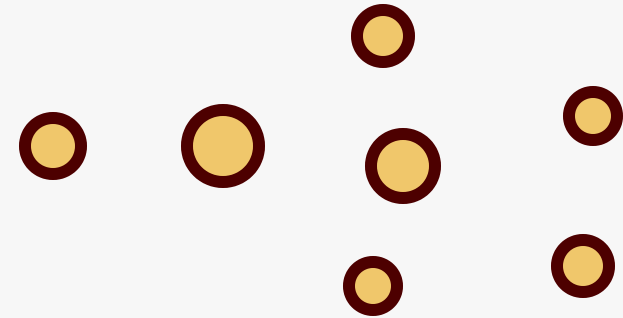


61.6

remove every
connection



Measurements only



61.8

Prediction did not get worse. The current GNN mostly used vessel measurements
—not their connections.

A negative result saved us from the wrong next step

1

Keep the useful signal

How vessels enhance over time is worth studying.

2

Stop adding static complexity

More graph types and more static layers were not answering the real question.

3

Learn the movie itself

The next model should learn contrast flow through time before predicting response.

A “no improvement” result is valuable when it tells us what to stop—and what to test next.

Next: learn contrast flow before treatment response

PLANNED NEXT EXPERIMENT — NOT YET A RESULT



~1,000 exams

Learn without treatment labels

With vs. without the map

Test whether connections help

Before vs. after pretraining

Test whether learning the movie improves response prediction

What should you remember?

1

Vessel contrast behavior carries modest—but uncertain—information about response.

2

Our first **graph neural network** **did not benefit** from the vessel connections.

3

That result redirected us toward **learning contrast flow** through time.

A simpler model won this round—and showed us exactly what to test next.